Scaffolded Task Instructions

Step 1: Study Scope & Participant Screener Formulation

1. **Target System & Task Selection**
    a. For the system selection I have chosen a website that I have done a study on prior. I will be using the Nothing brand's US website. This study will follow the process of navigating the phone pages. Both the purchasing path as well as tech specs comparison will be tested. I tried to select tasks that would be a good reflection of what users would likely have to do in real life scenarios. Both of the tasks cover pain points that were revealed in my prior lab assignment.
    b. Task 1: "Your phone carrier has told you that your area uses 5G on band n30. Using this website, find a phone that supports that band and add it to your cart. Please stop before entering any payment information."
    c. Task 2: "You have a friend who also likes this brand, but they want the best phone offered. Help them decide between the two models offered. Find the screen size, battery capacity, and front camera resolution for both phones. List the units in inches, mAh, and MP. You are only allowed to have one tab open at a time."
2. **Participant Screener Design**
    a. How often do you shop online?
        i. Several times a week, Once a week, A couple of times a month, Maybe once a month, A couple of times a year, Never (If never, exclude)
    b. Which phone brands have you owned or used in the past year?
        i. Apple, Samsung, Razer, Nothing, Google, Nokia, other (if Nothing, exclude, select all that apply)
    c. How old are you?
        i. <18, [18-25], [26-30], [31-39], [40-45], 46+, prefer not to say (if <18, exclude, used to balance)
    d. Do you use any accessibility features?
        i. Screen reader, Voice controls, Keyboard only navigation, Screen magnification, High contrast, other, none, prefer not to say (select all that apply, optional)

Step 2: Moderator Script & Informed Consent Protocol

1. **Ethical Participant Briefing**
    a. Hello and welcome to this study. You are free to pause, skip a task, or stop the test entirely at any point with no explanation, and no penalty. This is a test of a particular website, not a test of your performance. The screen, user voice, and notes that I take about your actions will be recorded during the test, with the screen and voice recordings being deleted by 1/1/2027. Participant names will not be recorded. All data will be stored securely in a password protected drive. Upon completion, anonymized data will be posted to a public GitHub repository, but all participant screen and voice recordings will remain on the locked drive. If you have any questions, feel free to ask them at any time, though questions asked during the test may be answered afterwards. At any point, up until the data is anonymized and posted publicly on 1/1/2027, you are able to request to have your data withdrawn. This test is expected to take about 45 minutes total. If you consent, please check the box for "agree".
2. **Concurrent Think-Aloud (CTA) Instructions**
    a. As you progress through the given task, please speak aloud your thought process, anything that you are looking for, anything that surprises you, what you think works well, or what you think doesn't work well. I will be present during this test, but I may hold off on answering questions that would assist with the task. If you need to pause, stop, or have the task re-read to you, I can do that. The purpose is to see how you would navigate this site if I were not here. I may remind you to speak aloud should you go quiet; this is for recording purposes, not because you are doing anything incorrectly. The idea is to narrate your actions, not explain them. For practice, you can take 30 seconds to narrate the process of adding someone's birthday to your calendar app on your phone.
        i. At start
    b. Please continue narrating your actions.
        i. If they go quiet
    c. Do whatever you think is best. Please proceed as if I was not here.
        i. If they ask a question
    d. Okay. Is this where you would like to stop? (move to SEQ & next task)
        i. When they finish a task
    e. We are evaluating how you would accomplish the given task.
        i. If they ask if they did something right.

f. That does look frustrating, but please remember that we are evaluating the site, not your performance. If you would like to take a break, we can do that.
    i. Visible frustration
g. Try checking the “Tech Specs” folder in the top left
    i. If the user asks for assistance.
h. Thank you for participating in the study. We were studying how participants would navigate this website. Please remember that you have until 1/1/2027 to request to have your data withdrawn from the study. In order to do so, please use the email address that was provided to you along with your assigned participant identification number. Please do not discuss the test with anyone else for the remainder of the week since we will be performing more tests. If you have any questions, feel free to ask them now.
    i. At end of study

3. **Task Scenario Handouts**
    a. Participant:
        i. Task 1
        ii. Your phone carrier has told you that your area uses 5G on band n30. Using this website, find a phone that supports that band and add it to your cart.
        iii. Please note: Stop before entering any payment information.
        iv. Tell the moderator when you are done.
        v. Task 2
        vi. You have a friend who also likes this brand, but they want the best phone offered. Help them decide between the two models offered. Find the screen size, battery capacity, and front camera resolution for both phones. List the units in inches, mAh, and MP. Tell your answers to the moderator.
        vii. Please note: Only use one tab or window.
        viii. Tell the moderator when you are done.
    b. Moderator:
        i. Participants will be given two tasks in a randomly chosen order:
            1. Task 1: Your phone carrier has told you that your area uses 5G on band n30. Using this website, find a phone that supports that band and add it to your cart.
            2. Rules: Stop before entering any payment information.
                a. Correct answer, Phone (3)
                b. Terminal action: When the phone is added to cart.

    c. Success scoring: 1 if participant checked at least one phone's 5G band and found n30 support and added the correct phone to cart without asking for help. 0.5 if they asked for assistance in finding the information. 0 if they failed to find the information, or if they added either phone without checking.
    d. Time limit: 5 minutes. Inform participant that time has been reached if not done with task. Move on to survey and next task.
    e. Sub-task checkpoints: Find tech specs, find 5G bands.
    f. Errors: User navigates to any page besides phone or phone selection pages. User clicks on a decorative UI element to try and gather information. User adds the wrong phone to cart.
3. Task 2: You have a friend who also likes this brand, but they want the best phone offered. Help them decide between the two models offered. Find the screen size, battery capacity, and front camera resolution for both phones. List the units in inches, mAh, and MP.
4. Rules: Only use one tab or window.
    a. User will tell them to you.
    b. Correct answer, 3 vs 4a: 6.67" vs 6.83", 5150mAh vs 5080mAh, 50MP vs 32MP
    c. Terminal action: When the participant says they are done.
    d. Success scoring: 1 if all numbers are correct. 0.5 if they asked for assistance in finding the numbers. 0 if they failed to find all of the required numbers. Record all numbers correct out of 6.
    e. Time limit: 15 minutes. Inform participant that time has been reached if not done with task. Move on to survey and next task.
    f. Sub-task checkpoints: find tech specs, find each of the 3 desired pieces of information
    g. Additional note: record how many times the participant has to travel from one phone's page to another
    h. Errors: If the user opens a second tab or window. User navigates to any page that is not a phone or phone

selection page. User records the wrong value for any of the details of the phones.

ii. You are only allowed to give the approved response if the participant asks for assistance.

iii. Please see CTA section for what to say during the study.

Step 3: Performance Metrics & Data Logger Sheet Design

1. **Quantitative Data Collection Architecture**
    a. Please see usability_logging_sheet
2. **Qualitative Observation Notes**
    a. Please see usability_logging_sheet

Step 4: System Usability Scale (SUS) Survey Instrumentation

1. **Standardized 10-item Instrument**
    a. https://docs.google.com/forms/d/e/1FAIpQLSelCKS5tdUrpVNGJJL7TH_4rCQUnPK3L-8w6-GY4ISkV54zaA/viewform?usp=dialog
2. **Scale Architecture**
3. **Scoring Algorithm Blueprint**
    a. Attached to form

Step 5: Accessibility-First Survey & Environment Audit

1. **WCAG 2.1 AA Instrument Compliance**

Criteria	WCAG SC	How I Tested	Result	Evidence
Automated Accessibility Audit	1.4.3, 1.3.1	Chrome DevTools, Lighthouse	94/100, 3 issues. Contrast in footer, no main landmark, heading order. All not fixable	Figure 1
Programmatic labels	1.3.1, 4.1.2	Chrome DevTools, Accessibility	Pass, uses ARIA	Figure 2
Keyboard navigation and visible focus	2.1.1, 2.4.7	Manually, using keyboard	Pass, logical order and arrows plus tab navigate whole page.	Figure 3

Text contrast >= 4.5:1	1.4.3	Chrome DevTools, color picker	Pass, contrast ratio of 16.48 on question text	Figure 4
Error messages announced (aria)	4.1.3	Chrome DevTools	Pass, error message has role alert	Figure 5

Post-Mortem Reflection

1. Why does the System Usability Scale (SUS) deliberately alternate between positive and negative phrasing across its 10 items, and what cognitive response bias does this structure safeguard against during user testing?
    a. Originally I had forgotten to calculate the -1 and 5- for the even and odd columns, and the result from my test entry came back as 0. After I changed the calculations to be correct, the result of my test entry turned out to be 50. This test entry was made by selecting the same column for every answer. Using this, I can infer that this system was put in place to prevent the user from just selecting every option in a column. It actually makes the user take the time to think about each question before responding. This prevents acquiescence bias since the user is not likely to select the positive option across all questions.
2. Describe how a moderator's subtle verbal cues or body language during a Think-Aloud session can inadvertently corrupt time-on-task and error-count data. How does your protocol script prevent this experimental bias?
    a. If the moderator makes any indications that the participant picks up on, they might be given a false sense of reassurance that they are doing something either correctly or incorrectly. The moderator could also respond in a way that might make the user feel better about their results even if they did poorly, meaning that the results of the survey might differ from what they would have otherwise recorded. My script uses neutral language to the best of my ability to ensure that the participant is not swayed towards either direction. The idea is to be as invisible as possible; only being there to record or provide any pre-approved assistance. My script has pre-recorded responses, but I can't control what the moderator themselves do outside of the script. They could still show some body language that might persuade the participant into taking or not taking certain actions.

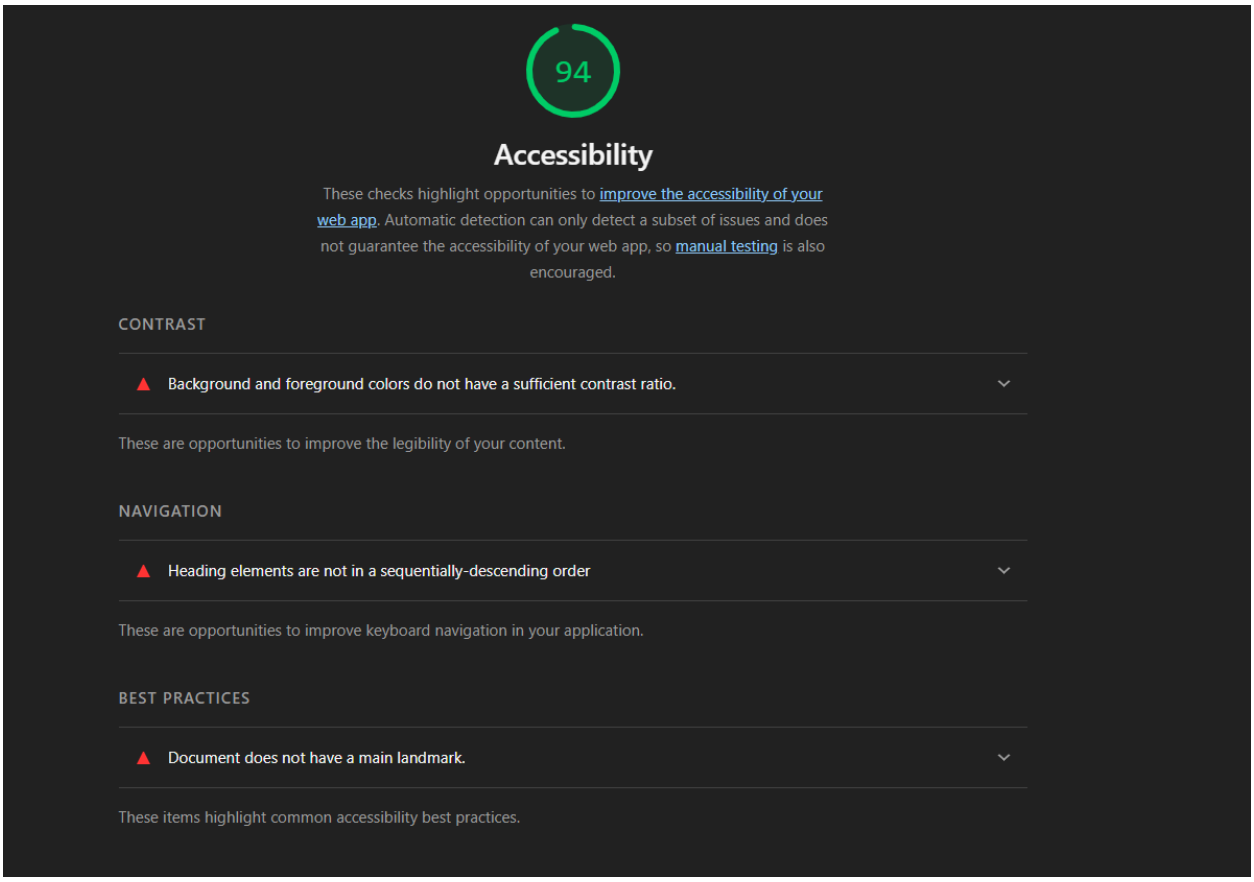


Figure 1. Lighthouse results.

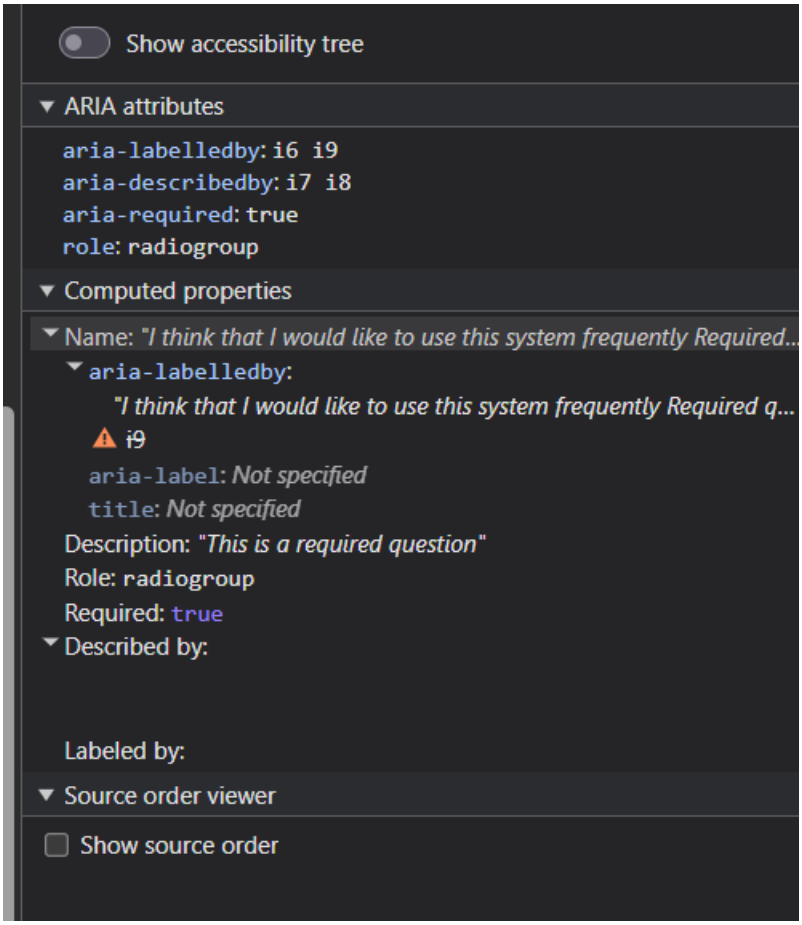


Figure 2. Aria name and role for radio button group.

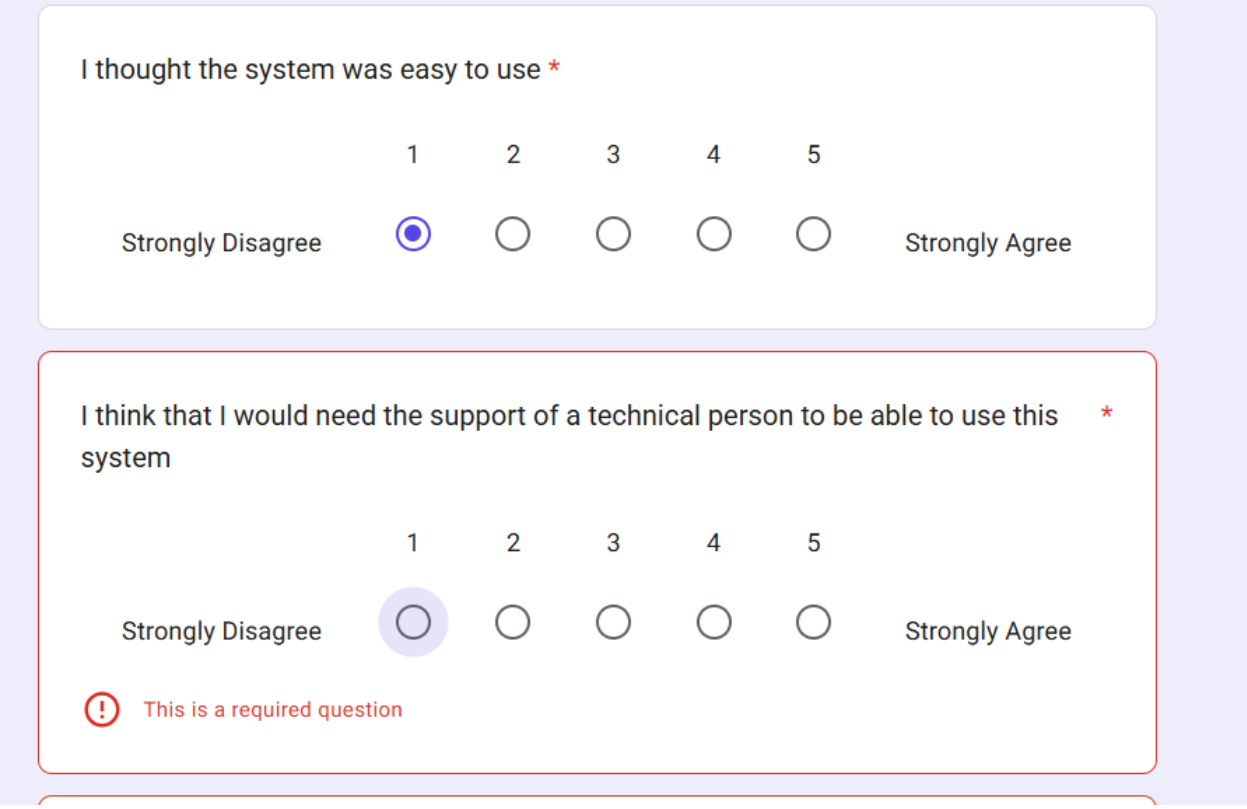


Figure 3. Visible focus.

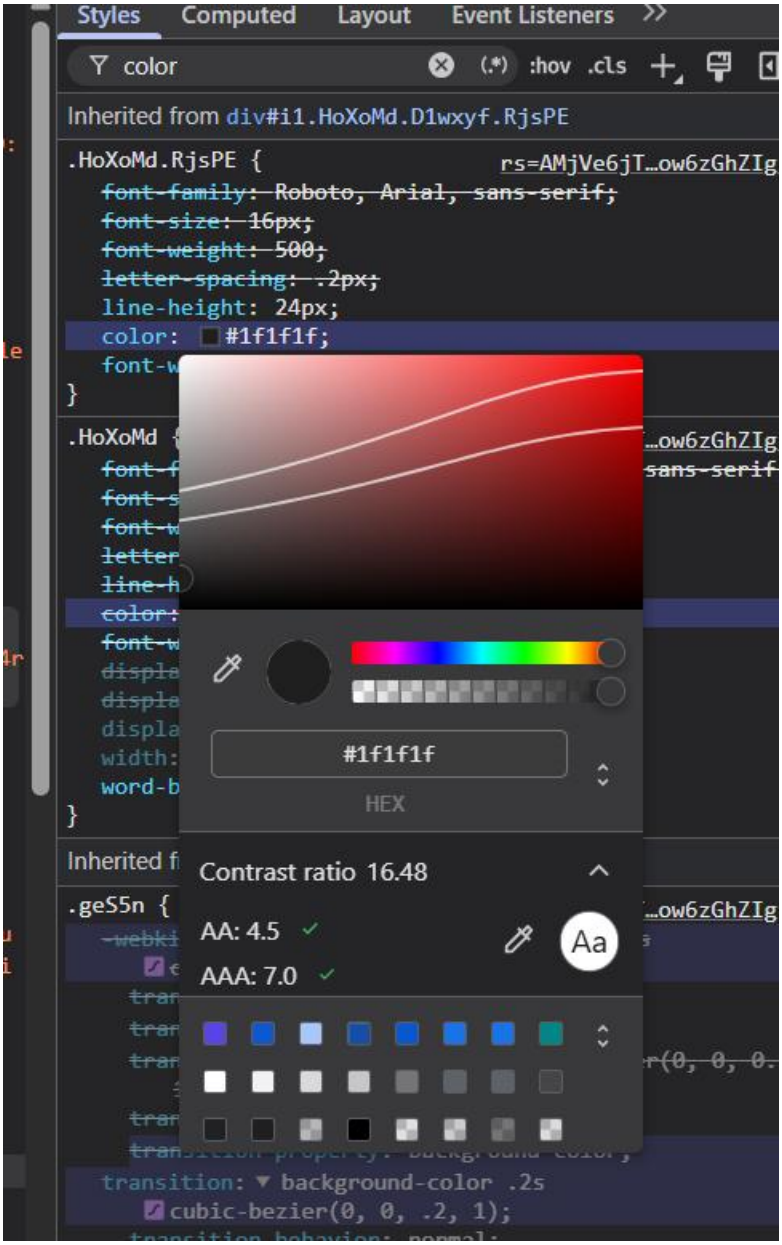


Figure 4. Text contrast ratio.

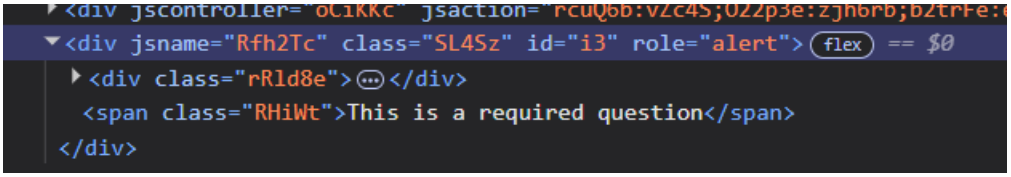


Figure 5. Alert text has alert role.